// loading workshop...

# Openclaw Workshop

$ build your personal agent

hosts Agentic_AI@uiuc × Build_illinois

deploy baadalAI

// host

# Agentic_AI@UIUC

# 300+

> members · #1 AI club on campus

```
const club = {
  mission  : "teach AI + agents",
  builds   : "open-source for uiuc",
  ships    : "workshops · projects · research",
};
```

// what_we_do

| | |
|---|---|
| > teach | LLMs, RAG, agents, evals |
| > build | tools used across UIUC |
| > host | weekly lectures + workshops |
| > ship | semester-long dev projects |
| > connect | research labs + startups |
| > grow | from curious → builder |

# ABOUT BUILDILLINOIS

- a community of the most ambitious builders, change-makers, and creatives on campus
- second path for students to work on their own ideas
- making founders and helping  founders of tomorrow

# WHAT BUILD HAS DONE

- hosted 1000+ people with summits, hackathons, retreats, firesides
- YC, Sequoia, Google, Afore, Prize Picks, Soma, ZFellows, Etched, and more
- 200-person hackathons with $120k+ in prizes
- private dinners + firesides with partners from 1517 Fund, Drive Capital, Afore, Doordash, Fanatics, and Metlife.
- organized a retreat sponsored by Google in Knox with 7-figure arr and Exited Founder

# WHAT BUILD HAS DONE RECENTLY

- UIUC's largest AI hackathon (250+ participants) with Respan YC W24 as a host, working with sponsors like AWS, Simplify, Lovable, and Red Bull
- 200+ person YC General Partner Pete Koomen x Respan founders fireside chat and a 15-person ZFellows founder roundtable
- panel with Fleetline YC S25 with founder Saurav Kumar

buildillinois

Fleetline

@uiuc

# APPLY TO BUILD? NO.

# NO MEMBERSHIP. NO REQUIREMENTS. JUST QR CODES.

slack

instagram

upcoming events RSVPs

(openclaw w/ agentic ai, gies-only copilot-backed buildathon, UT austin hook 'em hacks, cachehacks w/ foundry)

## GET INVOLVED:

- sundays: building + brainstorming + hang out
- next friday: OpenClaw Workshop w/ Agentic AI

// deployment_partner

# BaadalAI

> AI cloud deployment + monitoring

```
class baadal :
  def deploy  (app) → pick best cloud
  def monitor () → cost · traffic · perf
  def scale   () → auto up/down
  def secure  () → keys · tokens · logs
  def focus   () → "devs, not ops"
```

> powering every nanoclaw deploy today

// workshop deploy

$ **baadal deploy**

--target

**> hetzner**

cheap · eu

**> digitalocean**

fast · global

// why_vps

# why a vps?

> agents need to run 24/7. your laptop can't.

```
// your_laptop
class Laptop :
  def sleeps  () → "when lid closes"
  def dies    () → "battery @ 0%"
  def drops   () → "wifi goes out"
  def reboots () → "every update"
  def costs   () → "$1.5k+ hardware"
> NOT 24/7
```

```
// vps
class VPS :
  def runs    () → "never sleeps"
  def power   () → "always on"
  def network () → "100% uptime"
  def ip      () → "public · static"
  def cost    () → "~$5/mo. cheap."
> 24/7 · $5/mo
```

// providers

# pick a vps.

> two good ones. both work with baadalAI.

// option_01

## HETZNER

> german cloud · eu-based

| | |
|---|---|
| price | ~€4 / month |
| entry | CX22 · 2vCPU · 4GB |
| bandwidth | 20 TB unmetered |
| regions | DE · FI · US-east |
| ram | ECC · enterprise-grade |
| best_at | cheapest raw power |
| trade | fewer regions · eu-first |

**$ best value**

// option_02

## DigitalOcean

> dev-first cloud · global

| | |
|---|---|
| price | ~$6 / month |
| entry | droplet · 1vCPU · 1GB |
| bandwidth | 1 TB included |
| regions | 15+ worldwide |
| extras | k8s · spaces · docs |
| best_at | 1-click apps · tutorials |
| trade | pricier per gb |

**$ best experience**

// agenda

# ./today.sh

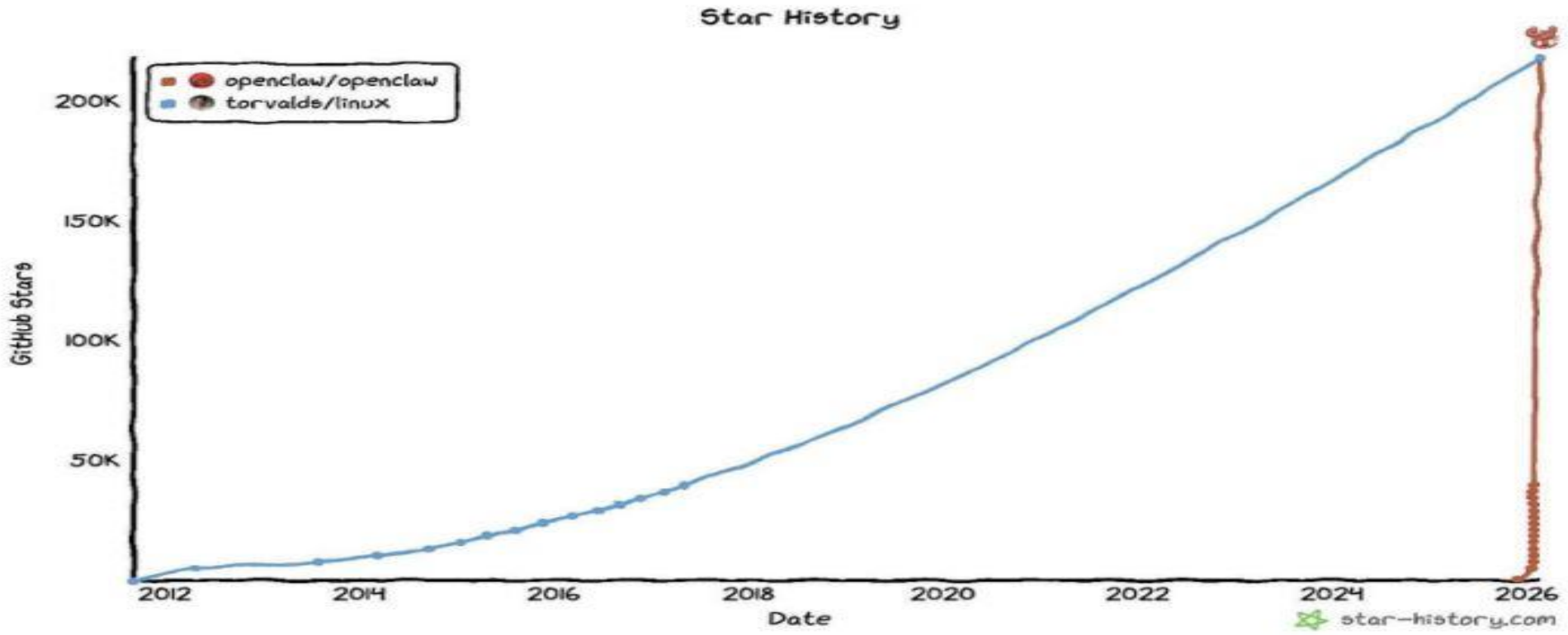
| | | | |
|---|---|---|---|
| 01 | demo | live nanoclaw | 10m |
| 02 | model | brain · skills · memory · loop | 15m |
| 03 | setup | repo · env · run | 25m |
| 04 | first | daily assistant | 40m |
| 05 | skills | add real powers | 55m |
| 06 | yours | personalize | 35m |
| 07 | debug | why agents fail | 20m |
| 08 | demo | showcase | 20m |

// foundation

# OpenClaw

```
const openclaw = {
  type: "agent framework",
  purpose: "see, think, act",
  defines: [ skills, tools, actions ],
  decides: when & how to use them,
  executes: "real tasks, not chat",
};
```

# OpenClaw Growth!!!




OpenClaw legitimately did overtake Linux on GitHub

// foundation

# OpenClaw Meetup in China!!!

// how_it_works

# 4 parts.

01

**brain**

llm

02

**skills**

tools

03

**memory**

context

04

**loop**

think → act

> user → think → tool → repeat → output

// the_catch

# heavy.

> great for prod. rough for a workshop.

- deps
- config
- slow boot
- deep stack

// meet

# Nanoclaw

> your personal AI operator.

| | |
|---|---|
| runs | locally |
| scope | single-user |
| setup | minimal |

nanoclaw =

**openclaw/child**

// vs

# openclaw vs nanoclaw

| dimension | openclaw | nanoclaw |
|---|---|---|
| setup | 30-60m | < 5m |
| deps | heavy | python + 1 key |
| boot | many parts | one cmd |
| scope | multi-user | single-user |
| for | production | learning |

> same model. faster ship.

// setup

# $ 5 minutes.

nanoclaw — bash

```
$ git clone github.com/agentic-ai-uiuc/nanoclaw
$ cd nanoclaw && python -m venv .venv
$ pip install -r requirements.txt
$ cp .env.example .env  # paste your API key
$ python agent.py

nanoclaw> hello
agent   > ready.
```

// first_agent

# daily_assistant

**input → think → output**

// prompt

```
"organize my day.
 summarize tasks.
 suggest next actions."
```

// response

```
1. ship workshop deck
2. reply to 2 emails
3. deep work 4-5pm
→ open deck.pptx
```

// skills

# pick 2-3.

- read_files()
- extract_tasks()
- web_search()
- plan_day()
- run_shell()
- write_code()

> do. don't just respond.

// prior_art

# Superpowers Plugin

> obra's skills framework — now in claude code marketplace

```
const superpowers = {
  author  : "obra · jesse vincent",
  type    : "agent dev workflow",
  skills  : "20+ composable",
  method  : "brainstorm → plan → ship",
  license : "MIT · free · open-source",
  install : "claude code marketplace",
};
```

> don't jump to code — **think first. ship in chunks.**

// core_commands

**/brainstorm**

refine rough ideas

**/write-plan**

break into chunks

**/execute-plan**

ship each step

> github.com/obra/superpowers

// prior_art/workflow

# the_workflow()

> don't write code — brainstorm → spec → plan → execute

01

## brainstorm

*// what are you really trying to do?*

> step back.

> no code yet.

> tease out the spec from conversation.

02

## spec

*// digest-sized chunks*

> show spec in chunks short enough to read.

> you sign off on each piece.

03

## plan

*// for a junior engineer*

> clear enough for an enthusiastic jr.

> red/green TDD.

> YAGNI · DRY.

04

## execute

*// subagent-driven*

> agents work tasks, review, continue.

> hours of autonomous work without drift.

> skills trigger automatically — **your agent just has superpowers.**

// make_it_yours

# ?

## what do you do every day
## that you could automate?

**students**

lectures · study plans

**builders**

debug · scripts

**everyone**

journal · plan · decide

// debug

# why they fail.

| | |
|---|---|
| ERR infinite loop | → add max_steps |
| ERR hallucinated tool | → ground with tool list |
| ERR bad output | → explicit format |

> design behavior, not code.

// debug

# THEME: Second Brain For Final Exams

> a **second brain** for your semester

// nanoclaw.ingest()

## stores + understands

> **notes**
lecture scribbles, handwritten

> **pdfs**
textbooks, papers, readings

> **lectures**
slides, transcripts, recordings

> **ideas**
half-baked thoughts, links

// why_it_works

## students already struggle with notes

> this just upgrades it.

// demo

```
upload(notes) → {
  summaries  // tl;dr of each chapter
  flashcards // spaced-rep ready
  quizzes    // test yourself
}
```

// showcase

# $ ./yours.py

> **show** what you built

> **share** the weird ideas

> **steal** from other people

// now go

# Build Build Build!!!

$ thanks to our partners

agentic_ai@uiuc

build_illinois

baadalAI